3. Bundle 1 — Instructions and Signal Walkthrough

RAT state at start: identity mapping (xi → pi). Free List starts at p32.

Slot	Instruction	rd	rs1	rs2	Type
ins0	ADD x1, x2, x3	x1	x2	x3	R-type
ins1	ADD x4, x1, x5	x4	x1	x5	R-type
ins2	ADDI x1, x1, 10	x1	x1	—	I-type
ins3	ADD x6, x4, x1	x6	x4	x1	R-type

3.1 Expected Signal Values (visible in GTKWave t=25 to t=35)

Signal	ins0	ins1	ins2	ins3
prd	32	33	34	35
prs1	2	32	32	33
prs2	3	5	0	34
old_prd	1	4	32	6

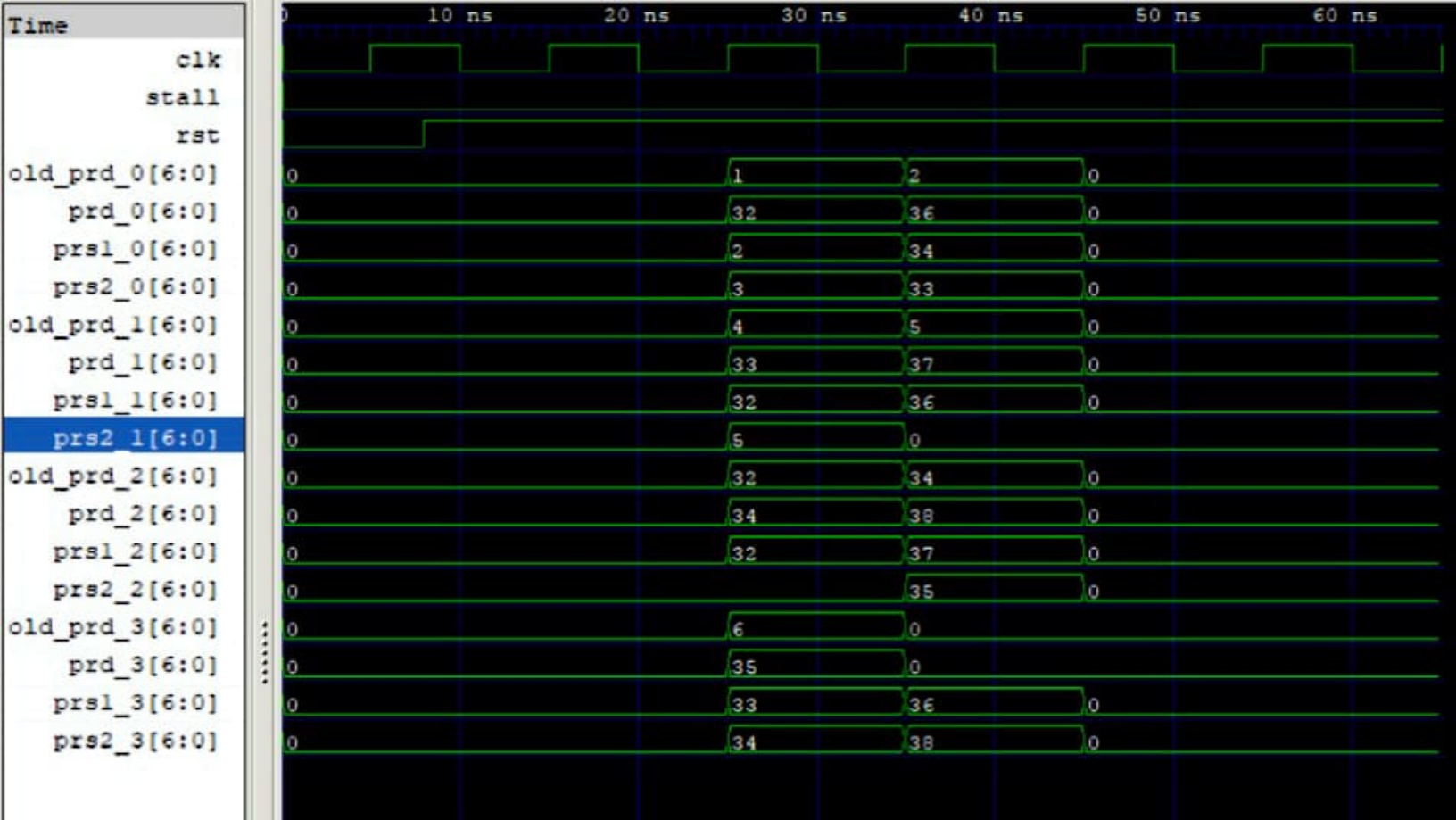
4. Bundle 2 — Instructions and Signal Walkthrough

RAT state entering Bundle 2 (committed after Bundle 1): x1→p34, x4→p33, x6→p35, all other registers still identity. Free List next available register is p36.

Slot	Instruction	rd	rs1	rs2	Type
ins0	ADD x2, x1, x4	x2	x1	x4	R-type
ins1	LW x5, 0(x2)	x5	x2	—	Load
ins2	ADD x1, x5, x6	x1	x5	x6	R-type
ins3	SW x1, 4(x2)	—	x2	x1	Store

4.1 Expected Signal Values (visible in GTKWave t=35 onwards)

Signal	ins0	ins1	ins2	ins3
prd	36	37	38	0
prs1	34	36	37	36
prs2	33	0	35	38
old_prd	2	5	34	0


Time
clk
stall
rst
old_prd_0[6:0]
prd_0[6:0]
prs1_0[6:0]
prs2_0[6:0]
old_prd_1[6:0]
prd_1[6:0]
prs1_1[6:0]
prs2_1[6:0]
old_prd_2[6:0]
prd_2[6:0]
prs1_2[6:0]
prs2_2[6:0]
old_prd_3[6:0]
prd_3[6:0]
prs1_3[6:0]
prs2_3[6:0]
0
10 ns
20 ns
30 ns
40 ns
50 ns
60 ns